

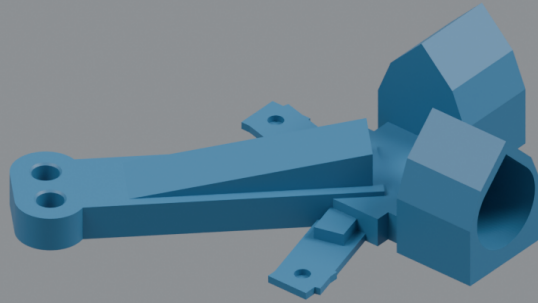
Front + rear guys / one revised top elbow

FIELDWORK / FPV RACING GATE

01

Two independent guys / one diagonal ear

Two 10.5 mm through-eyes in a 14 mm-thick ear. The load arm is 32 mm wide with a sloped root rib.



Two top/side magnetic pads replace the corner pad. PVC bore stays 33.5 mm, with the same 30 mm engagement.

Baseline documented and committed as 6a7fb47. New print: 102.57 g PETG / 3.95 hours. Two independent 10.5 mm through-eyes, two edge magnet pads, and the same 33.5 mm PVC sockets. Print two per gate, after testing one.

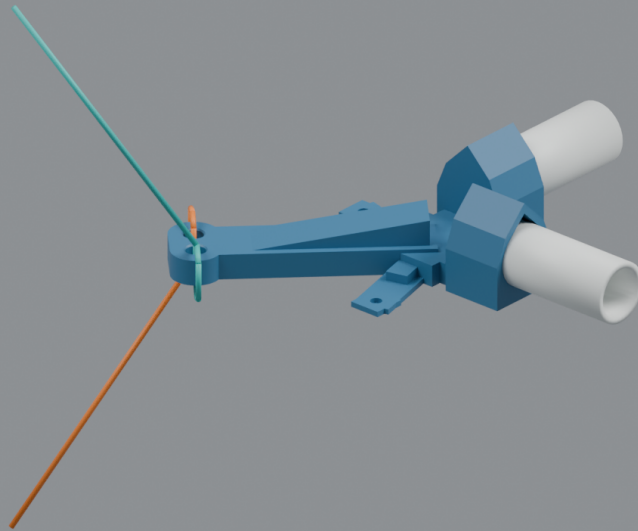
Two separate lines at the same corner

FIELDWORK / FPV RACING GATE

02

Orange to the front / teal to the rear

Separate eyes keep the two lines independently attachable. Both have 0.8 mm entry chamfers.



Loops illustrate threading only; use tested knots or suitable soft loops. No eye load or wind rating is assigned.

Each eye carries a separate attachment loop. Orange leads toward the front and teal toward the rear. The illustrated 3 mm loops clear the printed body; knots and physical loads remain to be tested.

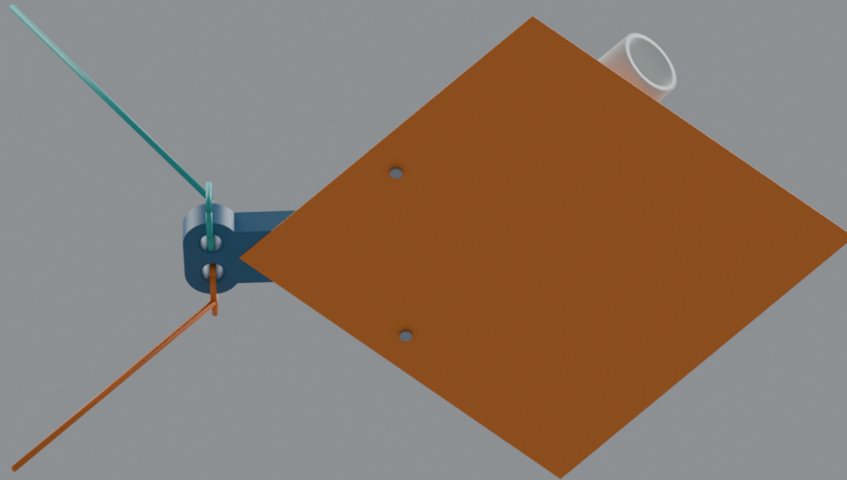
Clear the paper / move the magnets

FIELDWORK / FPV RACING GATE

03

Front guy access without puncturing paper

The diagonal ear projects beyond the paper corner. Each magnetic pad sits 18 mm inside one edge and 90 mm along it.



Only the small tie-down ear is exposed beyond the paper. The accepted PVC interfaces and all cut lengths remain unchanged.

The ear projects about 28.6 mm beyond the adjacent top/side paper edges. Both holes are fully accessible without piercing paper. Magnet centers are 18 mm inside the top/side edges and 90 mm from the corner.

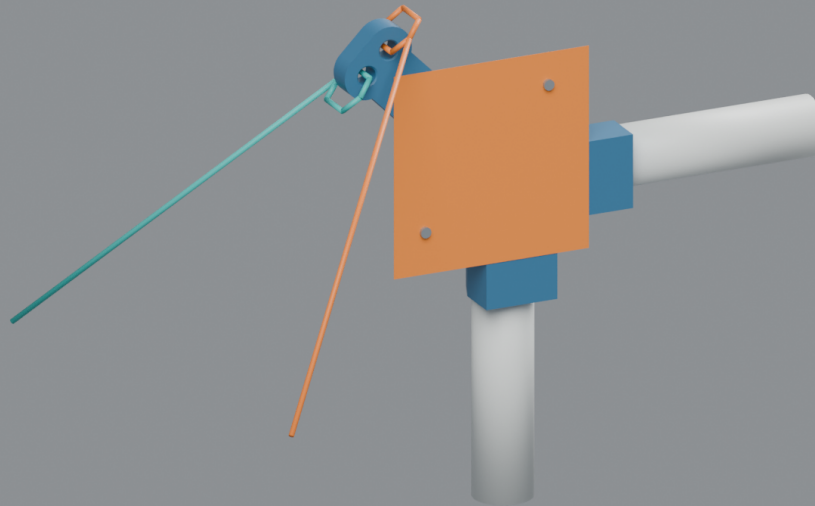
Install both guys together

FIELDWORK / FPV RACING GATE

08

Installed corner / front and rear support

One eye for each guy at each top corner. Both lines leave the face without a paper hole or conflict with a magnet.



Orange = toward the front. Teal = toward the rear. Ground endpoints are not shown or specified in this close-up.

The installed upper-left corner is shown with a paper cutaway. Both top corners use the same rotated part. Route the lines through their separate eyes and clear of the magnetic pads.

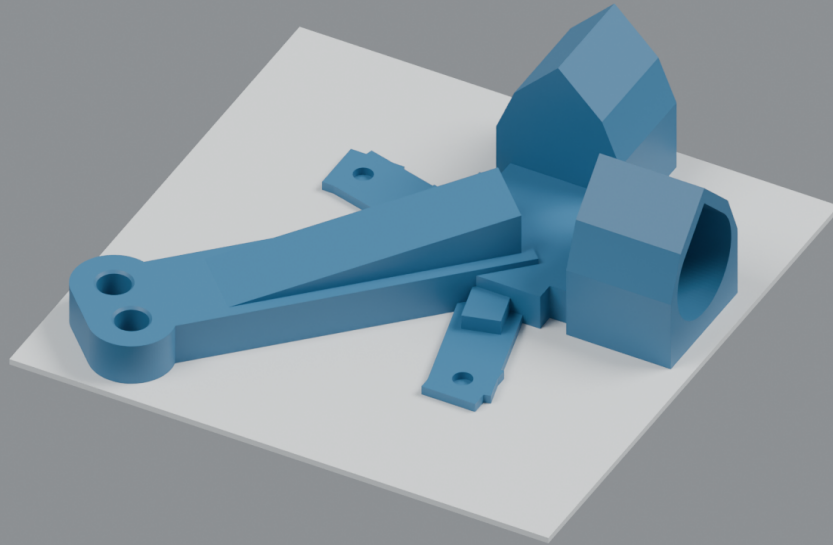
Print one on the A1 Mini

FIELDWORK / FPV RACING GATE

07

One support-free PETG print

Flat paper face and ear on the bed. Vertical tie-down holes need no roofs; PVC sockets retain their peaked roofs.



Print one, test both PVC sockets and both guy loops, then repeat for the second top corner.

168.614 x 168.614 x 53.345 mm. Supplied layout has at least 5 mm bed margin. PETG / 0.20 mm / four walls / 20% infill / no supports. Retain the supplied orientation.

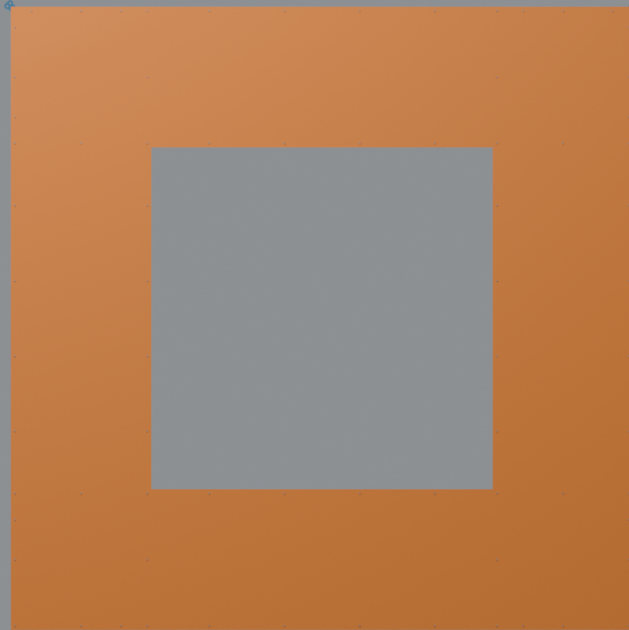
Same paper gate / two new top corners

FIELDWORK / FPV RACING GATE

05

Two revised top corners / same gate face

2700 mm square, 1480.8 mm opening, 609.6 mm bands. Magnet total is now 74 pairs / 148 individual magnets.



Five production part types and 64 pieces. Replace the two old guy elbows; keep the other parts and PVC cuts.

Single: 64 printed pieces in five types and 74 magnet pairs. Stacked: 104 pieces and 118 pairs. PVC cut lengths and paper dimensions remain unchanged.

Parts and production queue

Part	Single	Stacked
ELBOW	2	2
DUAL_GUY_ELBOW	2	2
TEE	8	12
CROSS	4	8
MAG_SLEEVE	48	80

Recipe	Single	Stacked	g/run	h/run
ONE_ELBOW	2	2	68.47	2.69
ONE_TEE	8	12	98.49	3.77
ONE_CROSS	4	8	132.59	5.07
ONE_DUAL_GUY_ELBOW	2	2	102.57	3.95
BATCH_MAG_SLEEVE	6	11	96.74	4.46
TAIL_MAG_SLEEVE_6	1	0	82.93	3.85
TAIL_MAG_SLEEVE_3	0	1	41.5	2.07

Single: 2.324 kg / 94.4 hours. Stacked: 3.690 kg / 150.3 hours. The two revised elbows add 50.36 g and 1.81 hours versus the committed tee design. Four additional individual magnets are needed.

Assembly and testing

Fit and thread both lines

1. Print one DUAL_GUY_ELBOW. Inspect the 0.4 mm magnetic skins, two vertical through-holes, diagonal arm and root rib. Clear loose strings and smooth any rough entrance edges without enlarging the PVC bores.
2. Fit two actual PVC offcuts and seat both sockets the full 30 mm, using insertion witness marks. The design bore stays 33.5 mm; the longer socket fit still needs its first physical check despite the successful sleeve print.
3. Thread a separate attachment loop through each eye and around its adjacent outside rim. Lead one guy toward the front of the gate and the other toward the rear. The orange/teal Blender lines show this routing; they are schematic loops, not a knot prescription. Choose appropriate secure knots/loops for the actual line and confirm knot clearance with both fitted simultaneously.
4. Use smooth cord or a protected connection at the plastic. Bare metal wire should not abrade directly against a printed edge. The modeled line diameter is **3 mm**; test the actual chosen cord or protected wire loop rather than treating this as a universal hardware fit.
5. Lay a paper corner over the fitting without a hole or notch. Align its two edges with the 18 mm magnet inset. Fit the two rear/front magnet pairs, and check that both cords can be connected and released with the paper in place. Bond rear magnets only after checking polarity and fit.
6. With the corner supported, try each guy independently, then both together, watching the elbow root and the pipe engagement marks. Check for visible flex that does not recover, whitening, cracks, cord-edge wear and socket withdrawal. Opposing guys do not remove the loads in the printed part. Repeat attachment/release and inspect again before committing to the second corner or an outdoor setup.

The digital routing check sampled **1046 points** on the illustrated loops and local tails against the actual printed mesh, with a minimum centerline distance of **5.049 mm** for 3 mm cord. The sampled paper crossings also clear the paper. Knots, flexible deformation and real line loads are not simulated. **This is a fabrication prototype with no assigned eye or wind load rating.** Ground support and anchor sizing remain separate installation work, particularly for the tall Split-S frame.

Replace the top corners in the gate

Install two DUAL_GUY_ELBOWs in place of the two old GUY_ELBOWs. The native part has PVC ports +X and +Y; the assembly maps rotate it 270 degrees at upper left and 180 degrees at upper right. The ears point outside the overall corners. All stop positions, PVC lengths and paper cuts remain identical.

Facing the gate, the new top-left magnet centers are **(18, H-90)** and **(90, H-18)** mm from the paper's bottom-left origin. Top-right centers are **(2700-18, H-90)** and **(2700-90, H-18)**. H is 2700 mm for single or 4790.4 mm for stacked. The updated Magnet_Positions.csv files include these changes; remove the old single corner position at each top corner.

Each top elbow now needs two magnet pairs instead of one. Totals become **74 pairs / 148 magnets single**, or **118 pairs / 236 magnets stacked**; **four more individual magnets** than the committed tee design. Paper roll usage, overlap seams and all other magnet positions are unchanged.

The gate still has **five required printed types, 64 pieces single / 104 stacked**. Single production estimate: **2.324 kg / 94.4 hours**, 23 plate runs. Stacked: **3.690 kg / 150.3 hours**, 36 plate runs. Count the sleeve already printed toward the BOM; full queues are from zero.

PVC requirements remain 16 short + 8 long cuts for single, or 24 short + 14 long for stacked, at **389.6 / 1560.8 mm**. Eight or fourteen 10-ft sticks at the supplied \$6 price. At an assumed \$20/kg PETG, PVC + PETG face-frame subtotals are **\$94.47 / \$157.81**, excluding magnets, paper, cord, adhesive and ground support.

Full print queue

Each CSV queue links to prepared A1 Mini PETG projects. Only ONE_DUAL_GUY_ELBOW is a new part/job; the other geometry matches the committed files. No print has been submitted.

Assembly and extension

The row-by-row assembly from the committed tee guide still applies, substituting the two new top elbows and new top magnet locations. Use checked PVC master pieces for repeated marking as recorded in the baseline; the optional jig is not required.

To extend upward, move the complete top row with both dual-guy elbows, its two tees, three pipes and sleeves. Add **4 tees, 4 crosses, 32 sleeves, 8 short PVC cuts and 6 long cuts**. Every existing pipe, fitting and populated magnetic attachment is reused. Two top elbows still provide four guys total: front and rear at each corner. Anchor endpoints and tension settings are not prescribed by the illustrative Blender views.

Verification and source

Checks cover connected manifold masters; A1 Mini plate bounds, clearances and support-free toolpaths; exact queues and node ports; unchanged non-guy STLs; matching magnetic pockets; full hole-mouth clearance beyond paper; sampled cord/body and cord/paper clearance; and reuse of all single-height parts on extension. The exposed diagonal ear is explicitly allowed in the front-projection check; other protrusions fail it.

Prusa's modeling guidance discusses how geometry and print orientation affect mechanical behavior. The arm dimensions and cord layout here are original prototype choices, not a strength rating derived from that source.

Rebuild with scripts/build_dual_guy_gate.py, slice_dual_guy_gate.py, validate_dual_guy_gate.py and package_dual_guy_gate.py. Baseline commit: **6a7fb47**. The new iteration is separate in output/dual_guy_gate; earlier printable parts and documentation remain intact.